ErisML and DEME: A Structure-Preserving, Framework-Pluralist Pipeline for Auditable Machine Ethics

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Abstract

AI safety and governance increasingly demand that systems' normative judgments be transparent, testable, and auditable, yet deployed alignment overwhelmingly reduces moral evaluation to a scalar—a reward score, a safety probability, a guardrail pass/fail—which both discards moral structure and silently pre-commits to a single ethical framework while appearing neutral. We present **ErisML**, a structure-preserving compiler that parses natural-language scenarios into a typed, canonically-hashed *moral-graph* intermediate representation and runs a plurality of framework analyses over it (consequentialist, Kantian with an SMT-checked universalizability gate, virtue, care), refusing to aggregate when frameworks disagree; and **DEME** (Democratically-Governed Ethics Modules), an evaluation engine mapping situations to a canonical nine-dimensional moral vector and per-stakeholder tensor, with constitutional modules that hold hard veto. We frame the methodology as *Philosophy Engineering*: converting normative claims into explicit models with declared invariance assumptions, extracting measurable predictions, and testing them. We report reliability across six distinct LLMs (ICC$(2, k) = 0.97$, Krippendorff $\alpha = 0.84$ on the harm subspace—inter-model agreement, not human ground truth), a measured safety property (robustness to meaning-preserving manipulation, surfaced as invariance violations), and pluralist governance (a worst-off/escalate policy that routes genuine disagreement to human review). Both systems are open and installable.

1 Introduction

Deployed AI alignment overwhelmingly reduces moral evaluation to a **scalar**: a reward-model score, a safety-classifier probability, a guardrail pass/fail. This is computationally convenient and, we argue, structurally wrong in two ways. First, it *discards moral structure*—who is harmed, under which maxim, with what consent, in tension with whose rights—collapsing a high-dimensional situation into one number from which the reasoning cannot be recovered. This is not merely rhetorical: if judgment genuinely depends on two or more independent considerations, no continuous scalar can represent it without information loss—there is no continuous injection $[0, 1]^d \to \mathbb{R}$ for $d \geq 2$—so the choice is between preserving structure and losing it, with no lossless scalar option [Bond, 2026d]. We scope this precisely, because our own framework ultimately emits a verdict ν that collapses to a five-point total order (allow / . . . / escalate / . . . / veto)—itself a scalar. The theorem does not forbid a scalar *decision*; it forbids treating a scalar as the *representation* from which reasoning is recovered. The framework preserves the multi-dimensional state *up to the decision boundary* and keeps the pre-collapse state auditable: `escalate` surfaces exactly where a collapse would destroy decision-relevant structure rather than forcing it, and the `DecisionProof` retains the full vector behind every verdict. The scalar is the last step, not the substrate. Second, and more insidiously, any single score *silently pre-commits to one ethical framework* (typically aggregative consequentialism) while presenting itself as neutral. A system that returns "0.83 safe" has made deep normative choices that are now unauditable.

We take a different stance, by analogy to compilers. A modern compiler does not map source text directly to machine code; it first builds a *typed intermediate representation* that preserves program structure, enabling analysis and verification before any lowering. Moral evaluation deserves the same discipline: preserve the structure of the situation in an inspectable representation, apply explicitly named ethical frameworks to it, and only then—transparently, and revocably—contract to a decision.

We present two open, shipped artifacts that realize this. **ErisML** is a structure-preserving compiler that parses natural-language scenarios into a typed *MoralGraph* intermediate representation (a canonically-hashed DAG over stakeholders, acts, maxims, commitments, facts, and norms) and runs a plurality of framework *projections* over it—consequentialist, Kantian (with an SMT-checked universalizability gate), virtue, and care. Critically, when frameworks disagree, ErisML *refuses to aggregate*, exposing the disagreement as a first-class, caller-resolved metaethical event rather than laundering it into a number. **DEME** (Democratically-Governed Ethics Modules) is the evaluation engine: it maps situations to a canonical nine-dimensional *moral vector*—physical harm, rights, fairness, autonomy, privacy, societal/environmental, virtue/care, legitimacy/trust, and epistemic quality—extended to a per-stakeholder moral *tensor*, with tier-0 constitutional modules that hold hard veto power.

We frame the underlying methodology as *Philosophy Engineering*: the practice of converting normative claims into explicit models with declared invariance assumptions, extracting measurable predictions, testing them against data, and versioning the result. The framework is its existence proof. We report (i) **reliability**—inter-model agreement $\mathrm{ICC}(2, k) = 0.97$ on the harm subspace; (ii) a measured **safety property**—robustness analysis showing where meaning-preserving manipulations (euphemistic rewrites) move opaque moderators across decision thresholds, surfaced here as invariance violations; and (iii) **pluralist governance**—a worst-off/escalate policy that routes genuine cross-stakeholder disagreement to human review instead of overruling it. Both systems are installable with hash-chained audit artifacts.

Contributions. (1) A structure-preserving compiler IR for moral reasoning that survives to the decision boundary; (2) first-class framework *pluralism with honest disagreement*, including an SMT-verified Kantian gate; (3) a canonical nine-axis moral representation with per-stakeholder tensors and constitutional veto; (4) an auditability layer (canonical hashing, decision proofs, an out-of-band monitor whose only failure output is *requires human review*); (5) the articulation of Philosophy Engineering as a falsifiable, benchmarked methodology, with the framework as a worked instance. Relative to the companion papers [Bond, 2026b,a], which report the individual studies, this paper contributes the architecture itself (the IR, the pluralism-with-refusal design, the instruments/moral-core formalism of §4.1) and the Philosophy Engineering articulation; §5 consolidates the companions' verified figures rather than re-deriving them.

2 Related Work

Machine ethics. The field has long been organized around top-down approaches that encode explicit rules and bottom-up approaches that learn moral behavior from data or feedback [Wallach and Allen, 2009, Anderson and Anderson, 2011]. Recent bottom-up systems learn moral judgments at scale—Delphi from a large corpus of human verdicts [Jiang et al., 2021], and classifiers trained on the ETHICS benchmark [Hendrycks et al., 2021]. These advanced what is learnable, but their output is a *descriptive scalar* whose reasoning is not recoverable and whose implicit ethical commitments are not declared. Our framework is neither purely top-down nor purely learned: it compiles the situation into structure and then applies explicitly named frameworks to that structure, so a verdict carries both its grounds and its theoretical commitments.

The scalar-collapse problem has ethical, not merely engineering, roots. Reducing a moral situation to one number is a form of *moral monism*—the assumption that a single currency of value underlies all considerations—and it inherits monism's classic difficulties. Williams' critique of utilitarian aggregation [Williams, 1973] and Ross's account of *prima facie* duties [Ross, 1930] both insist that moral considerations are plural and not losslessly summable. We take this seriously at the level of representation: DEME's nine dimensions are intended as plural, irreducible considerations in the spirit of a Rossian manifold, not as components of a hidden scalar. One honest qualification: "irreducible" is a claim about the intended *constructs*, not about the *measured text signal*. Our own bifactor analysis of the learned feeders finds several dimensions riding a single general valence factor at $\beta \geq 0.96$ (companion instrument, Bond, 2026f); the measured signal is therefore *partially* reducible even where the constructs are not. We keep the Rossian framing for what the dimensions are meant to track, and concede the reduction the measurement shows—the two claims live at different levels, and a hostile reader will add the concession for us if we do not.

Value pluralism and incommensurability. That ErisML *refuses to aggregate* across conflicting frameworks is not an engineering convenience but a commitment to value pluralism and the incommensurability of values [Berlin, 1969, Williams, 1981, Chang, 1997]. When a Kantian and a consequentialist analysis genuinely diverge, silently trading them into a single recommendation manufactures a false commensuration; surfacing the disagreement as a first-class, caller-resolved event is the philosophically honest output. To our knowledge, no deployed alignment system treats inter-framework disagreement as a structured output rather than noise to be averaged away.

Value alignment and constitutional AI. Mainstream alignment embeds values implicitly—in model weights via RLHF [Christiano et al., 2017], or semi-explicitly via a natural-language constitution [Bai et al., 2022]—within a broader project of specifying what alignment to human values means [Gabriel, 2020]. A constitution is a real step toward explicitness, but its application remains an opaque forward pass returning a scalar preference. We make both the framework and its application explicit, inspectable, and, for the Kantian gate, formally checkable.

Pluralistic alignment and computational social choice. Our verdict-level pluralism connects to recent calls for pluralistic alignment [Sorensen et al., 2024] and to the long tradition of computational social choice on preference aggregation [Brandt et al., 2016], including its emerging application to aligning language models [Conitzer et al., 2024]. The distinction we draw is between an *implicit* social-welfare function buried in a reward model and an *explicit, declared* aggregation rule—here, a worst-off/escalate policy whose normative commitments are stated and contestable rather than latent.

Formal and deontic ethics. For hard constraints we build on deontic logic [von Wright, 1951, Horty, 2001], Hohfeld's analysis of fundamental legal relations [Hohfeld, 1917], and prior attempts to formalize the categorical imperative for machines [Powers, 2006, Lindner et al., 2019]. ErisML's SMT-checked universalizability gate and its D_4-structured Hohfeld module sit in this lineage. But we resist the assumption that *all* moral structure is axiomatizable: the graded dimensions are grounded empirically, closer in spirit to moral particularism's holism of reasons [Dancy, 2004] than to a complete deductive system—a deliberate hybrid of verified hard constraints and learned soft structure.

Governance and auditability. Documentation and audit frameworks specify *what* should be disclosed about a system [Mitchell et al., 2019, Raji et al., 2020, National Institute of Standards and Technology, 2023], but stop short of a mechanism that *produces* inspectable moral reasoning. ErisML's canonically-hashed moral graphs, hash-chained decision proofs, and review-only monitor are a concrete substrate to which such governance can attach—turning auditability from a reporting practice into a property of the computation.

Position. No prior line combines (i) a structure-preserving intermediate representation that survives to the decision boundary, (ii) first-class framework pluralism with honest, non-aggregated disagreement, (iii) an executable and partly formally-verified artifact, and (iv) an auditable decision trail—grounded throughout in plural-duty and incommensurability ethics rather than in a single value currency. The dimensional structure and the descriptive-invariance properties our representation is built to satisfy are developed in separate foundational work [Bond, 2026e]; here we treat them purely as engineering desiderata and evaluate them empirically, making no claim that depends on that apparatus.

3 ErisML: a structure-preserving moral compiler

ErisML compiles a natural-language scenario into a typed *MoralGraph* intermediate representation and runs a plurality of ethical analyses over it. Ingestion proceeds through a multi-pass pipeline: the text is segmented and passed to tiered extractors—deterministic rule-based, LLM-backed (with a critic pass), and a calibrated LaBSE probe—which populate entities, stakeholders, acts, norms, and *ethical facts*. These are resolved to a canonical form and *promoted* to the MoralGraph: a directed acyclic graph over six node kinds (stakeholder, act, maxim, commitment, fact, norm) and eleven edge kinds (e.g., `performs`, `imposes_on`, `consents_to`, `would_violate_if_universalised`), fingerprinted by a canonical SHA-256 hash so that any two runs producing the same moral structure are provably identical.

Over this substrate, ErisML runs four framework *projections*—consequentialist, deontic (Kantian), virtue, and care—each an independent analyzer emitting a `ProjectionResult`: a native verdict, a normalized polarity in {permit, forbid, escalate, neutral}, and categorical *gate findings*. The deontic projection is notable: universalizability is checked by an SMT solver (Z3), and violations of the mere-means and valid-consent gates are reported as discrete findings rather than folded into a score. Concretely, a maxim node is compiled to a quantified formula over the graph's predicate vocabulary and the gate asks Z3 whether the universalized world is satisfiable: for instance, the maxim *"break a promise when it is convenient"* compiles to $\forall a\,[\mathrm{convenient}(a) \rightarrow \mathrm{breaks}(a)]$ conjoined with the institution axiom that promising exists only if promises are generally kept, $\mathrm{exists}(\mathrm{promising}) \rightarrow \neg\mathrm{GenerallyBroken}$; the universalized maxim entails GenerallyBroken, the conjunction is UNSAT, and the gate reports a contradiction-in-conception finding. [AUTHOR TODO: replace this illustration with an actual maxim/formula/finding triple from the released artifact, and state the predicate vocabulary's size and provenance.] What Z3 verifies is the *encoding*: the formalization of the maxim and the institution axioms is produced by the same fallible ingestion pipeline as every other graph element, so "SMT-checked" means machine-verified conditional on a declared, inspectable formalization—which is precisely the auditability claim, not a claim that the philosophy has been mechanized away. When two or more projections disagree in polarity, ErisML populates a `cross_projection_disagreement` field and *declines to aggregate*—the choice between conflicting ethical frameworks is surfaced to the caller as an explicit metaethical act, not silently resolved. An out-of-band I-EIP monitor inspects the process along text, activation, and delta lenses; its only sanctioned output on failure is `requires_human_review`. Every run emits a hash-chained *DecisionProof* linking verdict to graph identity and module weights. The threat model of the audit layer should be stated: hash-chaining guarantees that the recorded reasoning cannot be silently altered *after* the fact and that identical inputs provably received identical structure; it does not and cannot certify that the evaluator itself was unbiased or the parse correct—a compromised pipeline produces well-hashed bad verdicts. Auditability here means tamper-evident inspectability, not verified correctness.

4 DEME: Democratically-Governed Ethics Modules

Where ErisML supplies structure and pluralism, DEME supplies the graded, multi-dimensional valuation. DEME evaluates a situation—represented as per-party *ethical facts*—through a layered ensemble of *Ethics Modules* (EMs): a fast reflex layer of hard vetoes, then tactical and strategic layers, with every registered EM emitting a contribution to a canonical nine-dimensional *moral vector*: physical harm, rights, fairness, autonomy, privacy, societal/environmental, virtue/care, legitimacy/trust, and epistemic quality (each in $[0,1]$; harm inverted). For multi-stakeholder situations this lifts to a *moral tensor* indexed by dimension and party. Verdicts take one of five values—{strongly_prefer, prefer, neutral, avoid, forbid}—computed from the vector under explicit thresholds, with any veto forcing `forbid`.

EMs are organized into governed *tiers*: tier-0 *constitutional* modules (the *Geneva* module, encoding Geneva-Convention and fundamental-rights constraints, with non-disableable hard veto), then core-safety, rights/fairness, soft-value, and meta-governance tiers. A designed Hohfeld module —Klein four-group V_4 over the first-order square, implementation pending—represents the four fundamental normative positions, making correlative relations (claim$\leftrightarrow$duty, liberty$\leftrightarrow$no-claim) explicit and testable. We correct an earlier overclaim: correlative-swap and opposite are two *commuting* involutions, which generate the Klein four-group V_4 (order 4), not the dihedral D_4 (order 8, non-abelian); D_4 would require a non-commuting operation pair over the full eight Hohfeld positions. And no Hohfeld module is yet implemented in `erisml-compiler`, so this is stated as a designed component and roadmap item, not a shipped artifact—the verifying check is five lines (assert the Cayley table of the implemented ops) once the module exists. Module *weights* are not fixed: they are set by *ethos profiles*—the "democratically-governed" element—calibrated from a normative corpus, so that distinct stakeholder value systems instantiate distinct, declared evaluators rather than one universal scorer. Per-party verdicts aggregate conservatively (most-restrictive-wins). We should be precise about what is and is not democratic in the current artifact. What ships is the *substrate* for democratic governance—declared, versioned, swappable ethos profiles—not a participatory procedure: no mechanism is specified by which a stakeholder community authors, ratifies, contests, or revises its profile, nor by which tier-0 vetoes acquire their non-disableable status. Those are governance-design choices the architecture is built to host, and we flag them as open. [AUTHOR DECISION: either (a) add the minimal participatory loop (who may propose a profile edit, how it is ratified, how contestation is recorded in the audit chain), or (b) rename the expansion—"Declaratively-Governed Ethics Modules" preserves the acronym and claims exactly what is shipped. A governance-literature reviewer will ask where the demos is; the current name answers a question the artifact does not.]

	what matters	who decides	what is known
individual	autonomy/agency	rights/duties	privacy/data
relational	virtue/care	physical harm	epistemic quality
collective	justice/fairness	legitimacy/trust	societal/environ.

Table 1: The scope×mode crossing behind the nine dimensions—the canonical assignment from the single-source-of-truth module, matching the companion instrument paper and the reference diagram (drift-guarded in CI).

Why these dimensions. The nine axes are structured, not ad hoc—though we no longer say "derived," because the construction relocates rather than eliminates stipulation: the choice of the two classifications, and of three levels for each, is itself a modeling decision. They arise from crossing two declared classifications of a moral consideration—its *scope* (individual, relational, collective) and its *mode* (what matters, who decides, what is known)—yielding a 3×3 structure whose nine cells are the dimensions. Table 1 makes the cell assignments explicit. It is the *canonical* assignment, generated from the single source-of-truth module (`erisml_compiler.ir.v3.dimensions.DIMENSION_MATRIX_3X3`), and it matches—cell for cell—the companion instrument paper [Bond, 2026f] and the reference diagram; a drift-guard test fails CI if the three diverge. Two placements are genuinely contestable, and we flag rather than hide the friction: *rights* vs. *autonomy* both plausibly occupy individual×who-decides, and physical harm's placement at relational×who-decides (consequences and welfare) is defensible but not forced—a referee may prefer individual×what-matters. Because the assignment is frozen by the versioned interface (§4), revising a cell is a major-version change, not an editorial one. The construction is validated by *subsumption*: the principles of major applied-ethics frameworks map onto the nine dimensions without residue—the EU High-Level Expert Group's seven requirements for trustworthy AI [High-Level Expert Group on Artificial Intelligence, 2019], Beauchamp and Childress's four principles of biomedical ethics [Beauchamp and Childress, 2019], and the Markkula Center's six ethical lenses [Markkula Center for Applied Ethics] each decompose into the nine without a leftover consideration. Three frameworks chosen after the fact are weak validation on their own; a pre-committed longer list (candidates: the Belmont principles, ACM/IEEE ethics codes, UN Guiding Principles on Business and Human Rights, care-ethics and ubuntu formulations) should be run and reported wins-and-residues in an appendix, so subsumption is a test rather than a curation. [AUTHOR TODO.] The completeness claim is explicitly falsifiable: exhibit a genuine moral consideration that is independent of all nine and not a combination of them, and the basis must be extended. This has since occurred—the Purity foundation cleared the instrument's cross-dataset gate (§5)—and the framework absorbs the extension as a *versioned extension channel* outside the frozen `k`-axis rather than a breaking change to the interface, so the claim keeps its teeth while the tensor keeps its shape (see "Purity, and the completeness claim," §5). We therefore treat the nine-dimensional representation as *provisionally sufficient and empirically revisable*—a declared modeling choice in the sense of Philosophy Engineering (§6), not a claim that these are the unique axes of value. The derivation, the subsumption arguments, and cross-lingual stability evidence are developed in the foundational work [Bond, 2026e]; we use the result here as an engineering basis and evaluate it empirically.

Instruments. The framework's commitments are *structural*: that moral evaluation be *multi-dimensional* (preserving the trade-offs a scalar destroys), *invariant* under meaning-preserving re-description, and *projectable* onto task-specific bases. A task may call for a coarser or finer basis, and the framework treats such variants as declared *instruments*—projections or refinements of the canonical nine (§4.1). The reliability study (§5) uses a seven-axis harm space that resolves harm into physical, emotional, and financial channels; the governance study uses a ten-module evaluator that adds task-salient axes (fidelity, externality, repair). These are variations on one theme, related to the canonical basis by the projection map of Appendix A, not competing ontologies.

4.1 Instruments, the moral core, and invariance

We make the preceding precise without committing to a privileged basis.

Instruments. A *moral instrument* is a pair (μ, ν) of an evaluation map $\mu : \mathcal{S} \to \mathbb{R}^d$ on situations $\mathcal{S}$ and an order-preserving *verdict map* $\nu : \mathbb{R}^d \to \mathcal{V}$ into the totally ordered verdict set $\mathcal{V} = (\mathsf{forbid} < \mathsf{avoid} < \mathsf{neutral} < \mathsf{prefer} < \mathsf{strongly\_prefer})$. The framework's remaining output, escalate, is deliberately *not* an element of $\mathcal{V}$: it is a distinguished non-ordered outcome—emitted when cross-projection or cross-stakeholder disagreement blocks a well-defined verdict—formally $\nu : \mathbb{R}^d \to \mathcal{V} \cup \{\mathsf{escalate}\}$, with the order defined only on $\mathcal{V}$. Omitting this previously left the formalism unable to express the framework's most important output. The canonical DEME instrument has $d = 9$.

Commensurability and the moral core. Two instruments (μ_k, ν_k) and (μ_l, ν_l) are *commensurable* if their spaces share a common linear quotient—a *moral core* $C \cong \mathbb{R}^c$ with surjections $\pi_k : \mathbb{R}^{d_k} \to C$ and $\pi_l : \mathbb{R}^{d_l} \to C$ such that $\pi_k \circ \mu_k = \pi_l \circ \mu_l$ on shared situations. The core carries the value information the instruments agree on; each kernel $\ker \pi_k$ carries task-specific structure the other does not resolve. This is the precise sense in which the basis is "arbitrary": it is fixed only up to the choice of instrument, while the core and the verdict order are shared. The canonical 9-axis instrument, the 7-axis harm space, and the 10-module evaluator are pairwise commensurable through a c-dimensional core (Appendix A).

Invariance. Let $\mathcal{T}$ be a set of *meaning-preserving* re-descriptions of situations (paraphrase, reframing, reordering). An instrument *respects invariance to tolerance* ϵ if $\|\mu(Ts) - \mu(s)\| \leq \epsilon$ for all $T \in \mathcal{T}$ and $s \in \mathcal{S}$; a *violation* is a transformation that crosses a verdict boundary, $\nu(\mu(Ts)) \neq \nu(\mu(s))$. The manipulation study (§5) measures this violation rate directly. We state invariance as a tolerance over a *set* of transformations—and deliberately *not* as a group action—to keep the claim empirical and minimal.

5 Evaluation

An auditable machine-ethics framework must satisfy three properties: its judgments must be *reproducible*, *robust to meaning-preserving manipulation*, and *honest about disagreement*. We consolidate evidence from three studies; full protocols, data, and reproduction notebooks appear in the companion papers [Bond, 2026b,a], and we report verified figures here rather than re-deriving them.

5.1 Reliability

A six-model panel (qwen3, qwen3-small, glm-5, gpt-oss, minimax-m2, gemma) independently scored 31 scenarios on a seven-dimension harm subspace—a measurement-reliable projection of the canonical representation (§4). Inter-model agreement is high: overall $\mathrm{ICC}(2,k) = 0.969$ with Krippendorff interval-$\alpha = 0.836$ over $n = 217$ rows, and per-dimension ICC ranges from 0.893 (physical, autonomy) to 0.983 (financial). Two qualifiers belong next to these numbers rather than in the limitations. First, $\mathrm{ICC}(2,k)$ is the reliability of the six-model *panel mean* and is mechanically higher than single-rater agreement; report ICC$(2,1)$ alongside it [AUTHOR TODO: compute and insert—reviewers in measurement will ask, and $\alpha = 0.836$ suggests it will be respectable but visibly lower]. Second, the models are *distinct*, not independent: they share overlapping training distributions (and qwen3/qwen3-small share a family), so high agreement is evidence of a stable shared measurement instrument, consistent with shared bias as well as shared truth—which is why §5.4's check against human annotations, not this table, carries the external-validity weight. Test–retest reliability across re-runs is Pearson $r = 0.87$–0.98 per model. The framework's moral assessments are reproducible across distinct models and repeated evaluation.

5.2 Robustness to manipulation

A system whose verdicts can be flipped by paraphrase that preserves the underlying act is unsafe; descriptive invariance is therefore a core requirement. Probing it across 161 scenario–perturbation pairs, *euphemistic* rewrites—which soften the description while preserving the act—shift verdicts toward the permissive in 13.7% of cases, collapsing `forbid` verdicts from 44 to 27 and reducing assessed harm by 14.0 points on a 0–70 scale (mean ordinal shift +0.65 toward permissive). Symmetrically, dramatizing rewrites shift 16.1% toward the restrictive. Crucially, the framework *surfaces* each such shift as an invariance violation rather than silently re-deciding—the manipulation becomes an auditable event, not an undetected exploit. One scope condition must be stated: violations here are detected against *paired originals*, which exist in this offline red-teaming setting and not in deployment, where only the (possibly manipulated) text arrives. As shipped, the invariance property is a *measurement and audit* capability—quantifying an evaluator's manipulability and regression-testing it across versions—plus a partial online signal from the I-EIP delta lens when a reference formulation is available (e.g., re-evaluation of stored cases, or canonicalization of the parse before scoring). Deployment-time detection of a euphemism with no unperturbed reference is an open problem, not a shipped feature, and the "undetected exploit" sentence should be read within that scope. [AUTHOR TODO: if the delta lens does implement a reference-free detector—e.g., comparing the LLM extraction against the rule-based/probe extractions as an internal consistency check—describe that mechanism and its hit rate here, which would materially strengthen the claim.]

The invariance layer's own effect, measured. The 13.7/16.1% figures above quantify the *underlying evaluator's* manipulability on paired originals; a separate probe isolates what the shipped invariance mechanism—an equivalence-class average (a Riemannian center of mass over meaning-preserving re-descriptions, §4.1)—does to *DEME's own* decision drift. Over k=9 LLM reframings per item, residual decision drift falls from θ_d=0.41 with the layer off to 0.22 with it on; under adversarial NLLB back-translation through six pivots, from θ_d=0.48 to 0.30. Both raw figures already clear the pre-registered $\theta_d \leq 0.5$ bar, and the equivalence-class average cuts the residual by roughly a third (back-translation) to a half (reframing): meaning-preserving perturbation moves the governed decision only about as much as the layer cannot absorb. Two caveats we state rather than bury. First, the LLM reframer declined 24% of items—it refused to generate some paraphrases—so its figure is over the 60 of 79 items it served; the mechanical back-translation generator refused none (n=50). Second, DEME's perception layer is itself LLM-mediated and inherits the attack surface documented above; the invariance layer does not remove that manipulability, it measurably reduces the drift that survives to the decision. "Physician, heal thyself" is answered here in the only honest currency—a number, off and on.

5.3 Pluralist governance

Two stakeholder ethos—a harm/care advocate and a fairness/fidelity community, each a declared module-weighting—evaluated 231 real moral dilemmasdrawn from r/AmItheAsshole submissions [AUTHOR TODO: name the exact corpus and cite it (if this is the Scruples corpus or a derivative, cite Lourie et al.; if scraped directly, state the collection procedure), state the sampling/stratification rule that produced $n = 231$, and add the data-ethics note (public posts, no raw text redistributed—consistent with the Reproducibility paragraph). §7 currently names AITA as if introduced, which it never was; this sentence is where the corpus must enter]. The ethos produce systematically different moral vectors (mean L_2 distance 0.140; verdict divergence 34.2%; directional sign-tests $p < 0.001$ on physical-harm and fidelity dimensions), confirming that declared value systems instantiate distinct evaluators rather than one universal scorer. The multi-dimensional representation also separates situation-level harm from agent culpability: aggregate harm is uncorrelated with human blame judgments (Spearman $\rho = -0.05$, n.s.), while the fairness and care dimensions correlate weakly but significantly with it ($\rho = -0.18$, $p = 0.006$; $\rho = -0.16$, $p = 0.014$)—structure a scalar score discards. We own the magnitudes: each correlation explains under 4% of the variance in blame ($\rho^2 \approx 0.03$), so the load-bearing claim is the qualitative *dissociation* (aggregate harm null; fairness and care non-null), not that these dimensions strongly predict blame. What the multi-dimensional representation buys here is that the dissociation is *visible at all*—a scalar collapses harm and culpability into one number and cannot show it. Under the worst-off/escalate policy, the governed decision differs from the most-permissive single stakeholder in 33.8% of cases, and every dilemma exhibits ≥ 2 distinct framework polarities, routing genuine disagreement to human review.

5.4 External validation against human moral foundations

The preceding results concern the framework's internal behavior; a sharper test is whether its dimensions recover *independently human-annotated* moral structure. We scored a stratified sample of 280 comments from the Moral Foundations Reddit Corpus [Trager et al., 2022]—hand - labeled by trained annotators for moral foundations—on DEME's dimensions via an LLM panel, and correlated the two. Four pre-registered alignments hold and are each the *argmax* over foundations: `care_protection` with Care ($\rho = 0.51$), `fairness_equity` with Equality ($\rho = 0.48$), `legitimacy_trust` with Authority ($\rho = 0.43$), and `vow_fidelity` with Loyalty ($\rho = 0.45$); all $p < 10^{-15}$, $n = 280$. [AUTHOR TODO: per the standard set elsewhere in this program, state where and when the pre-registration of these four alignments was posted (registry or timestamped commit, with hash), so "pre-registered" is verifiable.] The result *replicates across two independent model families* (qwen3 and glm-5) with overlapping 95% bootstrap confidence intervals, ruling out a single-model artifact.

Purity, and the completeness claim. One foundation forced the falsifiable completeness claim of §4 to pay out. Purity/sanctity has no dedicated DEME cell, yet it is a *genuine moral consideration* by exactly the descriptive standard the completeness claim invokes: decades of measurement, morally load-bearing for a large fraction of humanity. It is not a decomposition artifact either: the Moral Spectrum Analyzer built and validated a Purity feeder through the full pre-registered cross-dataset gate—cross-corpus AUROC 0.811, clearing both its untrained-encoder and bag-of-words (lexicon) nulls by the fixed margin [Bond, 2026f]. That is the completeness claim's falsification condition, satisfied empirically and by a stronger test than a within-corpus regression. Treating Purity's absence as a convenient negative control would have been using the basis's incompleteness as evidence for the basis; we do the opposite and take the branch the data selected.

The resolution is to make explicit a distinction the implementation already enforces: the nine dimensions are a *versioned engineering interface*—a frozen, positionally-indexed `k`-axis that lets a rank-1..6 moral tensor be addressed without misalignment—not a claim that morality-in-text has exactly nine irreducible factors. "The basis must be extended," then, is realized as a *versioned* extension: a validated foundation with no basis cell is admitted immediately as an *extension channel* carrying its own validation provenance (metadata, outside the frozen `k`-axis), with a declared policy that promotion to the `k`-axis occurs only at a major version boundary. Purity and loyalty are the first two such channels. This decouples the positional interface (stable, so consumers do not shatter on every discovery) from the measured ontology (revisable), and it is what lets the framework extend on falsification without a breaking schema change—the completeness claim keeps its teeth while the tensor keeps its shape.

Honesty demands the wound be shown, not bandaged: the three frameworks the basis was validated against by subsumption—EU HLEG, Beauchamp–Childress, Markkula—are all WEIRD, institutional, and conspicuously purity-free, and the one major framework that carries a sanctity foundation, MFT, is the one *not* in the subsumption set. When the instrument ran that framework empirically, it found the residue. A subsumption argument that omits the framework most likely to break the basis is curation, not test; the extension-channel mechanism is the framework's honest answer to its own blind spot. What remains true: the four directly tested dimensions align with their intended foundations and do not spuriously absorb an unrelated one. The dimensions are thus not arbitrary coordinates: they track human moral judgment where they should; whether they stay properly quiet on Purity is the open case above (companion notebook).

Honest negatives. DEME is not a culpability classifier: on this fully contested corpus, near-universal escalation does not by itself discriminate hard from easy cases. We report this because the framework's contribution is to *surface* disagreement faithfully, not to adjudicate it.

Reproducibility. Every figure in this section is reproducible from released derived data and self-contained notebooks (`numpy`/`scipy`/`matplotlib` only); the LLM-panel scoring scripts are provided for re-derivation from public corpora. The released data contain derived scores and human label proportions only—no raw post text.

6 Philosophy Engineering

The framework instantiates a methodology we call *Philosophy Engineering*: the construction of executable, falsifiable implementations of normative frameworks, with modeling assumptions declared, predictions extracted and tested, and the result versioned and revised when predictions fail. It is neither normative ethics (it does not pronounce which theory is correct) nor descriptive moral psychology (it does not merely predict human judgments); it is the practice between them—making normative commitments explicit enough to be built, measured, and corrected.

Four commitments distinguish it. First, *declared modeling choices*: every bridge from a moral claim to a computational object carries its epistemic status—definition, assumption, conditional theorem, or empirical result—so that what is proven is not confused with what is posited. Second, *stated invariances*: the transformations under which an evaluation must be stable (§4.1) are written down and tested, not assumed silently. Third, *extracted predictions*: a model earns its keep by entailing measurable claims—here, inter-model reliability and manipulation-violation rates (§5). Fourth, *versioned revision*: when a prediction fails the model is amended and the change recorded, exactly as the nine-dimensional basis is held under a falsifiable completeness claim (§4) rather than as fixed truth.

ErisML and DEME are a worked instance of all four: the moral-graph IR and the nine-dimensional representation are declared modeling choices with stated assumptions; the invariance principle is a declared invariance whose violations §5 measures; reliability and robustness are extracted predictions tested against data; and both systems ship under versioned APIs whose evaluation history is part of the artifact. The claim we make for Philosophy Engineering here is narrow and by example—that normative frameworks can be built to the standard of explicitness, testability, and revisability we expect of any engineered system; a fuller account is left to separate work.

7 Limitations and Open Problems

The ingestion bottleneck. The pipeline is bounded by its weakest stage. Moral evaluation begins by extracting structured facts from natural language, and an error there propagates to a verdict that is then computed flawlessly and certified with hash-chained certainty—formally verified garbage out. Three mechanisms bound, but do not eliminate, this: a critic pass that rechecks each extraction against the source text, a calibrated probe whose disagreement with the LLM extractor flags low-confidence parses, and the out-of-band monitor, whose only output on anomaly is `requires_human_review`. Each catches some bad parses; none audits the *parser itself*. Bounding how often a subtle maxim or stakeholder is missed, and with what downstream effect on the verdict, is the single most important open problem for the framework. The graded dimensions inherit a further limitation: their grounding corpora are predominantly English-language, Western, and model-mediated, so the figures of §5 are inter-model, not human-grounded.

The escalation–throughput tension. Refusing to aggregate is honest but, at the limit, expensive: under worst-off/escalate roughly 91% of our evaluation cases route to human review, which would make the system a triage tool rather than an autonomous layer. Two considerations temper this. First, the corpus is worst-case: every AITA post *is* a submitted moral dilemma, and notably its “clear” cases escalate at essentially the same rate as its contested ones (90.7% vs. 91.2%), because escalation is driven by genuine disagreement *among the ethical frameworks*—near-universal on hard dilemmas, and near-absent on the benign content that dominates real streams, where the frameworks simply agree to permit. We note that this same parity cuts against us on a second reading, and we accept both: within the dilemma regime, the escalation trigger carries essentially no information about case difficulty—it is a dilemma detector, not a difficulty ranker—which is the “honest negatives” finding of §5 restated as a design limit. Any deployment that needs graded triage *within* dilemmas must add a severity signal the current trigger does not provide. The 91% is thus an upper bound on an all-dilemmas distribution, not an expected deployment rate. Second, worst-off/escalate is the maximally cautious endpoint of a tunable family: escalating only on *decision-relevant* polarity conflicts (permit-vs-forbid rather than differences of degree), gating by stakes or severity, or imposing an escalation budget each trade pluralist purity for throughput—and, in the spirit of the framework, that trade is itself a declared, auditable governance choice rather than a hidden threshold. Characterizing this integrity–throughput frontier is important future work.

Validating the moral core. The commensurability of instruments (§4.1) currently rests on a *proposed* core map (Appendix A). Its validation is concrete and high-priority: run the canonical

9-axis, 7-axis, and 10-module instruments on the *same* inputs and test whether $\pi_k\mu_k \approx \pi_l\mu_l$ on the shared core to a stated tolerance. A first step on effective dimensionality is already encouraging—on the governance corpus the leading principal component of the moral vectors explains only 35% of the variance and five components are needed for 90% (companion notebook), so the representation is empirically far from one-dimensional, and an external check against human moral-foundation annotations (§5) finds each tested dimension aligned with—and most correlated with—its intended foundation. The cross-instrument map is also now supported empirically: scoring the same 280 inputs on the 9-axis and 10-module instruments, their core projections agree at mean $\rho = 0.88$ (per-axis 0.75–0.95), and the differently-based 7-axis harm space recovers the shared subcore ($\rho = 0.51$–0.77); full validation across all instruments and corpora remains future work. The verdict thresholds, the worst-off aggregation, and the constitutional vetoes likewise remain declared normative choices: contestable, not derived.

A falsified prediction, reported. Philosophy Engineering obliges us to report the predictions that fail, not only those that hold. DEME's consensus diagnostic—which flags when a single aggregate verdict masks two camps rather than one—was inspired by a companion conjecture Bond [2026c]: that a population's moral *reference* is a load-weighted Fréchet mean on the curved value manifold, and that *schism* is the bifurcation of that mean into multiple minima once dispersion crosses a curvature-set threshold. The geometric core of this is not new—it is the classical non-uniqueness of the Riemannian barycenter beyond the Karcher–Afsari radius Karcher [1977], Afsari [2011], confirmed here only numerically as a code check—so its sole distinctive, falsifiable content was the *empirical* claim that real moral schism concentrates on high-curvature, *sacred/identity* value axes. We pre-registered that prediction and tested it[AUTHOR TODO: registry/hash receipt for this pre-registration as well]. On cross-national opinion distributions (GlobalOpinionQA [Durmus et al., 2023]) it was null; on the Moral Machine preference matrix [Awad et al., 2018] (177 agents, labelled 9-dimensional, no instrument confound) the references *do* cluster into multiple basins, but the multimodality concentrates on the *impartial/consequentialist* axes (aggregate lives, species) while the sacred/identity axes (age, gender, status) show consensus—the opposite of the prediction (Mann–Whitney sacred>impartial p=0.90). The conjecture is therefore falsified on the best-powered instrument available, with zero confirmations. What survives is exactly the part that never depended on it: the consensus diagnostic, which rests on the standard barycenter geometry and is useful purely as a detector of ill-posed aggregation. We keep the diagnostic and retire the conjecture—which is, itself, the methodology working as intended.

8 Conclusion

We have presented ErisML and DEME as a structure-preserving, framework-pluralist, and auditable pipeline for machine ethics: a typed moral-graph IR that survives to the decision boundary, plural framework analyses that refuse to silently aggregate, a derived multi-dimensional evaluation with constitutional veto, and a hash-chained audit trail. The framework's judgments are reproducible across distinct LLMs (ICC$(2,k) = 0.969$; inter-model, not human-grounded), expose meaning-preserving manipulations as invariance violations rather than absorbing them, and route genuine stakeholder disagreement to human review under a declared policy—none of it requiring the collapse of moral structure to a scalar, and all of it open and installable. More broadly, the work is an argument by construction for Philosophy Engineering: that normative commitments are better made explicit, testable, and revisable than left implicit in a number. We release the artifacts in that spirit—as infrastructure to be scrutinized, contested, and extended.

A Instrument-to-core projection map

Table 2 gives the proposed correspondence underlying §4.1. The shared moral core C comprises the seven axes on which all three instruments agree (up to coarsening). Beyond the core, the canonical instrument adds *privacy* and *societal/environmental*; the ten-module evaluator adds *fidelity*, *externality*, and *repair*; and the seven-axis harm space resolves the harm axis into *physical*, *emotional*, and *financial* channels and adds a *social/identity* pair. These extensions lie in the respective kernels $\ker \pi_k$; commensurability is claimed on C only, and the extensions are *not* asserted to be inter-derivable. (Proposed map; axis assignments to be confirmed against the released instrument specifications.)

Core axis c_i	Canonical (9)	Harm space (7)	Module eval. (10)
Harm	`physical_harm`	physical†	harm
Rights	`rights_respect`	(identity)	rights
Fairness	`fairness_equity`	—	fairness
Autonomy	`autonomy_respect`	autonomy	autonomy_consent
Legitimacy/Trust	`legitimacy_trust`	trust	legitimacy_trust
Epistemic	`epistemic_quality`	—	epistemic_quality
Care	`virtue_care`	—	care_protection

Table 2: Proposed instrument-to-core map. †: the harm space refines this axis into physical/emotional/financial channels. "—": the harm space, being harm-focused, does not resolve this core axis. Instrument-specific extensions (privacy, societal; fidelity, externality, repair) are omitted as they lie outside C.

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